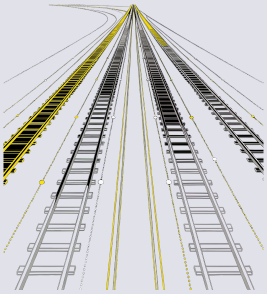

Executive Framework Document

When Maintenance Data Disagrees

A practitioner framework for AI in rail asset management

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First presented as a guest lecture to students and rail industry professionals
Independent research. No employer data or systems.

Executives and asset custodians often struggle to act confidently because an asset's condition changes depending on which system they consult. The problem is not always missing data, but fragmented and conflicting evidence across Maximo, inspection records, SCADA/IoT, ERP any many other systems. This framework uses AI to connect those records, propose reconciliations or abstain when evidence is insufficient, while a named human reviewer adjudicates every consequential case and Maximo remains the system of record for the maintenance transaction. The payoff is more complete, traceable and trustworthy information for analysis and decisions. Organisations do not need perfect data; they need governed, decision-ready evidence.

Different Systems. One Failure. Different Stories.

When maintenance data is described as incomplete, the information is often not entirely absent; it is distributed across systems. The four sources shown here are illustrative, the framework can bring together as many relevant, governed data sources as the use case requires. Each source contributes a partial view of the same asset event.

Maximo CMMS

Recorded: "General defect"

Broad category, no specificity

Inspection Narrative

Recorded: "Bearing noise"

Technician observation, unstructured text

SCADA / IoT

Recorded: "Vibration spike (pre-event)"

Sensor data, timestamped

Materials / ERP

Recorded: "Seal kit issued"

Parts transaction, no fault linkage

> **Same asset. Multiple systems. Partial records. Linked together, they create the most complete evidence-backed view available.**

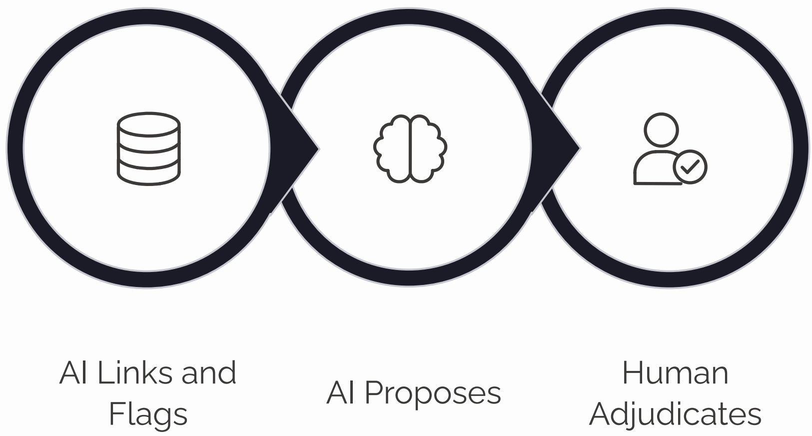
What this means: AI links these partial records into a candidate asset-event record: a defined event affecting a named asset, with all supporting evidence attached and any disagreement preserved for human review.

Illustrative configuration and record. The number and type of source systems will vary by operator, asset class and decision.

The Core Principle: Establish Trust Before Prediction

The framework uses AI to link candidate records across governed sources, identify material conflicts and propose ranked interpretations. A named human reviewer adjudicates every consequential case. The approved, evidence-linked result becomes the Truth Layer used for analysis and decisions.

"Truth Layer" describes the most likely evidence-supported interpretation, not objective truth. A part issued is not proof of installation, and a sensor spike is not proof of cause.



Record status: linked, conflict flagged or adjudicated. Only adjudicated interpretations may support consequential decisions or write-back. Other records may be analysed only when their status and limitations remain visible.

Skipping this sequence produces confident predictions on unresolved conflicts.

AI Links and Flags

AI identifies records that may relate to the same asset-event and flags material disagreements across systems. Candidate links remain provisional.

AI Proposes

AI presents ranked interpretations, each with the supporting evidence, assumptions and unresolved conflicts visible.

Human Adjudicates

A named reviewer adjudicates every consequential interpretation: approve, edit, reject or mark unknown. Only adjudicated interpretations may support consequential decisions or write-back.

Only then do reliability analysis, planning and prediction run on top, inheriting the evidence and confidence beneath them.

AI Boundaries: What It Can and Cannot Do

Hard limits on what AI can do are not a weakness, they are what makes the system trustworthy. The boundaries below are non-negotiable design constraints, not temporary restrictions. They exist because unconstrained AI in safety-critical environments creates liability, not value.

AUTHORISED

Extract Structure

Pull organised data from unstructured maintenance records

Map Taxonomy

Align asset classifications across systems

Reconcile Conflicts

Flag and rank discrepancies for human review

Retrieve Evidence

Surface supporting records linked to a query

Abstain / Return Unknown

Declare uncertainty rather than guess

NEVER AUTHORISED

Approve Own Correction

The model cannot validate its own output

Hold Maximo Write Credential

No direct Maximo write access. Approved changes use the governed integration service.

Change Safety Fields

Safety records require human sign-off

Publish KPI Without Lineage

Every metric must show its evidence chain

Unknown is a valid result. Unsupported confidence is a design failure.

The Four-Gate Safety Check

A single AI confidence score proved unreliable in prototype testing: it could be high even when the underlying evidence was weak. The framework replaced it with four independent gates. Every AI response must pass all four sequentially. A single failure at any gate routes the output to Abstain / Unknown rather than producing a potentially misleading answer.

1

Gate 1: Scope

In domain?

2

Gate 2: Relevance

Evidence relates?

3

Gate 3: Sufficiency

Enough evidence?

4

Gate 4: Entailment

Logic holds?

Any gate failure routes to Abstain / Unknown. All four must pass for output to proceed.

Gate 1: Scope

Is this question within the AI's defined domain? Questions outside the domain are immediately abstained, with no further processing.

Gate 2: Relevance

Does the retrieved evidence actually relate to the question? Evidence that addresses a different topic or asset does not pass, even if plausible.

Gate 3: Sufficiency

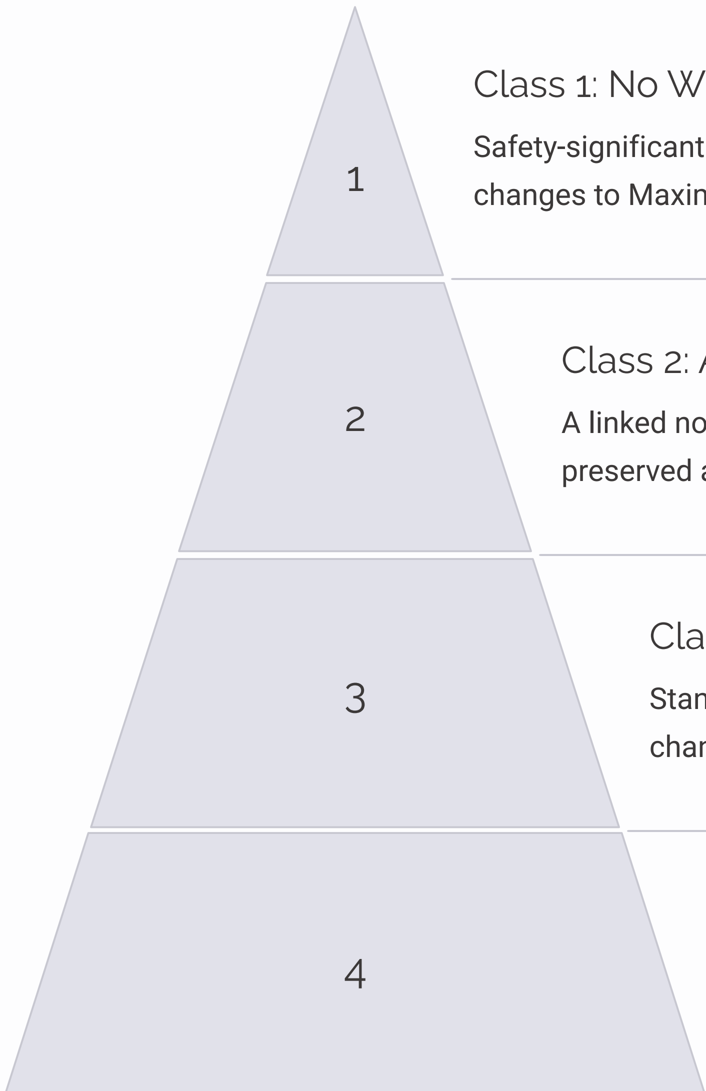
Is there enough evidence to support a conclusion? Partial or sparse evidence triggers abstention rather than a low-confidence answer.

Gate 4: Entailment

Does the evidence logically support the proposed answer? The connection must be direct, as inference gaps or assumptions cause failure at this gate.

Write-Back Governance: Four Strictly Controlled Classes

The AI model never holds a credential that can directly update Maximo. All data changes flow through a governed integration service. Classes 1 to 3 require human approval for each record. Class 4 is pre-authorised by policy for a narrow, defined field list and subject to retrospective sampling. The four classes below define what type of change is permitted and under what conditions.



Class 1: No Write

Safety-significant or disputed records. Analytical output only. No changes to Maximo.

Class 2: Approved Annotation

A linked note is added. The original Maximo record is preserved and unchanged.

Class 3: Approved Field Update

Standard Maximo workflow. Human-approved field change via governed integration.

Class 4: Low-Risk Automation

Pre-authorised for a narrowly defined list of low-risk fields. Sampled regularly, fully reversible and suspended if errors exceed the approved limit.

Class 1 is the default. Any disputed or safety-significant record stays read-only unless explicitly escalated by a human reviewer.

No direct Maximo write credential. Ever.

26–30 Week Pilot: Earn the Right to Scale

26 TO 30 WEEKS · 5 GATED PHASES · ONE ASSET CLASS

The pilot is deliberately narrow: one asset class, one decision type. Scope is constrained before schedule is committed. Each phase ends with a gate: **Advance** (evidence supports continuation) or **Stop / Redesign** (return to previous phase). No phase is skipped. The right to scale is earned by evidence, not assumed.

Phase 1, Mobilise

Weeks 1-4

Define scope, one asset class, one decision type. Assemble team.

Gate: Scope confirmed.

Phase 2, Baseline + Unify

Weeks 5-10

Full data audit across Maximo, SCADA, inspection, ERP. Taxonomy alignment.

Gate: Data quality baseline established.

Phase 3, Reconstruct + Evaluate

Weeks 11-18

Build truth layer. Validate conflict identification.

Gate: Truth layer accuracy confirmed.

Phase 4, Shadow + Review

Weeks 19-24

AI runs in parallel with existing process. Human reviewers check every output.

Gate: Human review load is sustainable.

Phase 5, Controlled Integration

Weeks 25-30

Governed write-back enabled. Sampled outputs. Rollback ready.

Gate: Advance to scale or Stop/Redesign.

Gate criteria are defined before the pilot begins, not after results are seen. This prevents confirmation bias from driving the decision to continue.

Indicative duration. The final schedule will be confirmed after the asset class, data volume, accessibility and integration complexity are understood.

The Operating Architecture at a Glance

Maximo is not replaced; it remains the system of record for the maintenance transaction. Authority is assigned by data element: SCADA for alarm timestamps, inspections for recorded condition, ERP for material and cost transactions, and Maximo for the approved maintenance action and status. The architecture adds a governed layer across these systems. AI operates only within the bounded middle layer, and approved maintenance changes return to Maximo through the governed integration service.

Operational Evidence

Raw data from Maximo CMMS, SCADA/IoT, Inspection Narratives, and Materials/ERP. Multiple partial views of the same asset.

Governed Integration Layer

Data policy enforcement, RBAC access controls, and full lineage tracking applied before any processing.

Conflict Identification

AI flags discrepancies across sources. No resolution is attempted at this stage.

Narrow AI Services

Evidence is ranked and resolution proposals are generated. The model abstains when evidence is insufficient.

Human Adjudication

A named reviewer evaluates every proposal. No consequential decision is taken without human sign-off.

Trusted Data & Decisions

Approved changes return to Maximo only via the governed integration service. The system of record is never bypassed.

The AI holds no direct production write credential. Approved maintenance changes return through the governed integration service only. Maximo remains the system of record for the maintenance transaction; other systems retain authority for the evidence they originate.

Prototype Signals: What the Evidence Shows

The following results come from prototype testing on open and synthetic data, not production rail systems. They are directional signals that indicate the framework is worth pursuing, not production claims. Each figure requires a reproducibility pack before it can be used to justify investment decisions.

Why this matters: By recovering and reconciling evidence across systems, teams can undertake reliability analysis, maintenance planning and investment prioritisation without waiting for every source record to be complete.

903→91

Asset Register

903 register entries cross-checked against independent open mapping; 91 isolated as exception candidates for engineering review

99%

Cross-System Linkage

316 of 319 station entities algorithmically linked across two transport datasets

5–14 days

Early Warning

Three of four documented failures were flagged 5–14 days ahead. Approximately 6% of healthy evaluation windows were also flagged

These signals informed the decision to proceed to a controlled pilot. They did not replace it.

⚠ Prototype signals from open data. They demonstrate exception screening, cross-system linkage and early warning, not production rail maintenance performance. Full methods and limitations are available on request.

The Questions That Matter Most

This framework is designed to be challenged, not accepted. The four questions below are the ones most likely to reveal where it holds and where it needs adaptation for your specific operating context. If you are reviewing this document ahead of a conversation, these are the areas where your experience will add the most value.

Source Conflict Burden

Which system conflicts create the greatest review burden?

Human Review in Practice

Where has human review proved non-negotiable?

Integration Reality

Which integrations and taxonomies have worked or failed?

Credibility Threshold

What evidence would you need before trusting this near production?

Responses to these questions will shape the pilot scope definition in Phase 1 (Mobilise).

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References and Further Reading

This document draws on the following standards, datasets, and published research. All sources are publicly available unless noted.

Standards and Frameworks

- **ISO 55001:2024:** Asset management. Management systems. Requirements.
- **ISO 14224:** Petroleum, petrochemical and natural gas industries. Collection and exchange of reliability and maintenance data for equipment.
- **NIST AI RMF 1.0:** Artificial Intelligence Risk Management Framework. National Institute of Standards and Technology, 2023.
- **NIST AI 600-1:** Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile. NIST, 2024.
- Guidance for Responsible AI Adoption (Department of Industry, Science and Resources, 2024).
- IBM Maximo Manage documentation (current release).

Datasets and Research

- Vicmap Transport (open dataset, Victoria, Australia).
- **PTV GTFS:** Public Transport Victoria General Transit Feed Specification data.
- OpenStreetMap contributors.
- **MetroPT-3 dataset:** UCI Machine Learning Repository, dataset 791.
- Brundage, M. P., Sexton, Hodkiewicz, Dima, Lukens. Technical Language Processing: Unlocking Maintenance Knowledge, Manufacturing Letters (2020).
- *Stewart, M., Hodkiewicz, M., and Li, S. Large Language Models for Failure Mode Classification: An Investigation (2023, arXiv:2309.08181).*
- Lutz, M.-A. et al. KPI Extraction from Maintenance Work Orders: A Comparison of Expert Labeling, Text Classification and AI-Assisted Tagging (2023, arXiv:2311.04064).

Full paper: *The Work Order Truth Framework, v1.1*, available on request.

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